



Department of Information Technology

Academic Year 2025 - 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.Tech. IT

Course Code & Title: CS3551 & Distributed Computing

Name of the Faculty member (s): Mrs. A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Lamport's Algorithm
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO6
- **Activity Chosen:** Learning by Teaching
- **Justification:**
Learning by Teaching is a powerful strategy for understanding Lamport's Algorithm in Distributed Computing. It helps students deeply grasp the concepts of logical clocks, event ordering, and causality among distributed systems. By explaining how Lamport timestamps establish a partial ordering of events, learners reinforce their conceptual clarity and develop critical thinking. Teaching peers encourages them to analyze message passing, synchronization issues, and consistency — key skills in real-world distributed environments.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - A topic "Lamport's Algorithm" was assigned to Mr.C.A.Parasuram one week before the presentation.
 - He prepared the presentation and delivered the session on 03.09.2025.
 - The topic was presented for about 15 minutes, with a demonstration example.
 - He explained the concept of logical clocks, event ordering, and the "happened-before" relation, using message diagrams and a sample distributed system with multiple processes.
 - The student demonstrated how Lamport timestamps can be used to maintain event ordering in distributed systems without relying on synchronized physical clocks.
 - Real-world examples like message ordering in distributed databases and synchronization in cloud services were used to illustrate practical applications.
 - The implementation of the Learning by Teaching activity is shown in Figure 1.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	3	3	2	2	2	2	3	3	3	2	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
	3	3	3	2	2	2	2	3	3	3	2	3
Justification for correlation	Students learn the core principles of Lamport's logical clocks and how they maintain event order across distributed processes.	They develop analytical and problem-solving skills by simulating message passing and identifying the correct event sequence.	Students enhance their algorithm design skills by applying Lamport's method to practical synchronization problems.	They strengthen their research abilities by exploring extensions like vector clocks and comparing synchronization techniques.	Students gain experience using simulation tools and pseudocode implementation to test algorithm accuracy.	Collaborative learning helps them work as a team while presenting, discussing, and verifying results.	They improve communication and presentation skills by explaining technical content clearly to peers.	Students develop project management abilities in organizing presentations and coordinating algorithm examples.	Continuous learning is encouraged as they stay updated with new developments in distributed systems and synchronization models.	Students apply data analysis and system modeling to understand timing dependencies among distributed events.	They learn to integrate emerging technologies like cloud synchronization and blockchain consensus protocols with Lamport's principles.	Students can implement Lamport's algorithm using open-source distributed system simulators for scalability and reliability testing.

• **Images / Screenshot of the practice:**



Figure 1: Learning by Teaching Handled by Mr.C.A. Parasuram

❖ **Reflective Critique**

❖ **Feedback of practice from students about the LBT:**

- Gained clear understanding of event ordering and Lamport timestamps.
- The peer teaching method made the complex concept easier to visualize and remember.
- Examples of message ordering and logical time updates enhanced analytical skills.
- Improved confidence in explaining synchronization algorithms and debugging event ordering problems.

❖ **Benefit of the practice:**

- Deep understanding of event sequencing and causality in distributed systems.
- Strengthened logical reasoning through timestamp calculation exercises.
- Better teamwork and peer learning through discussions and Q&A sessions.

❖ **Challenges faced in implementation:**

- Difficulty in visualizing concurrent events and partial ordering in large distributed systems.
- Required more hands-on examples and simulation tools to reinforce the concept.
- Some students needed extra guidance in differentiating Lamport clocks and vector clocks.

References:

- <https://effectiviology.com/protege-effect-learn-by-teaching/>
- <https://www.geeksforgeeks.org/lamports-logical-clock-algorithm-in-distributed-system/>

Signature of Faculty Member

HOD